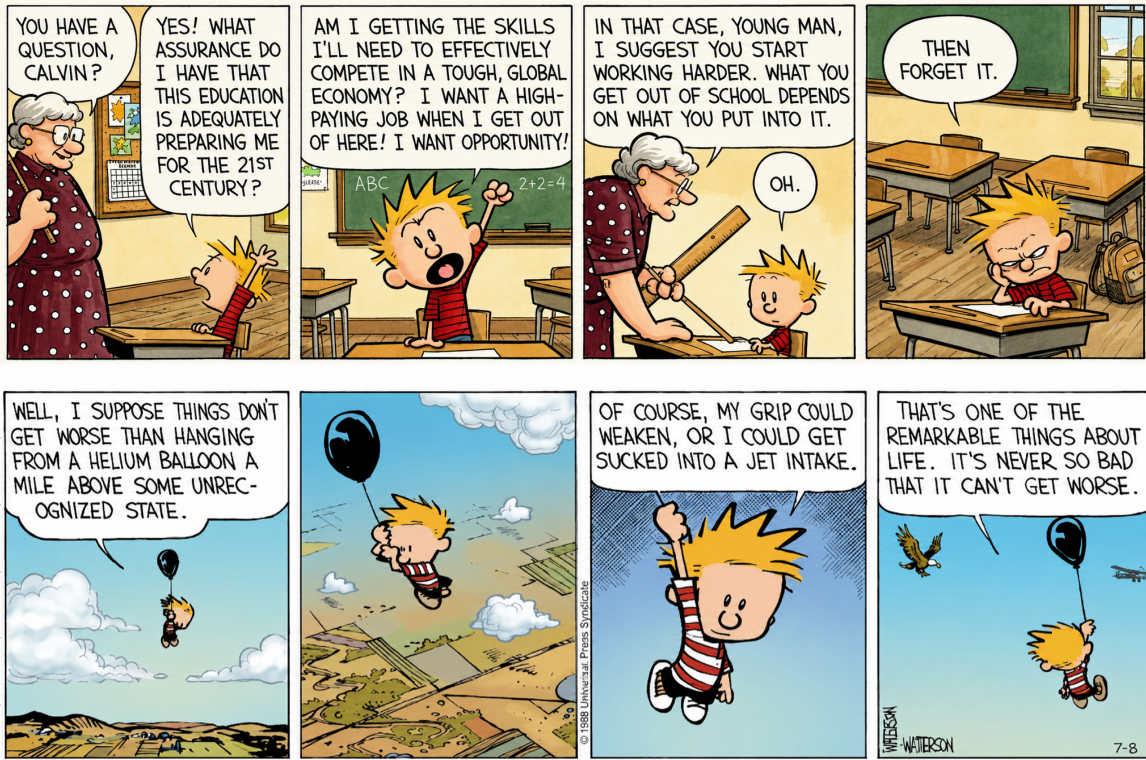


REMINDER :)

Please get Pen and Paper.

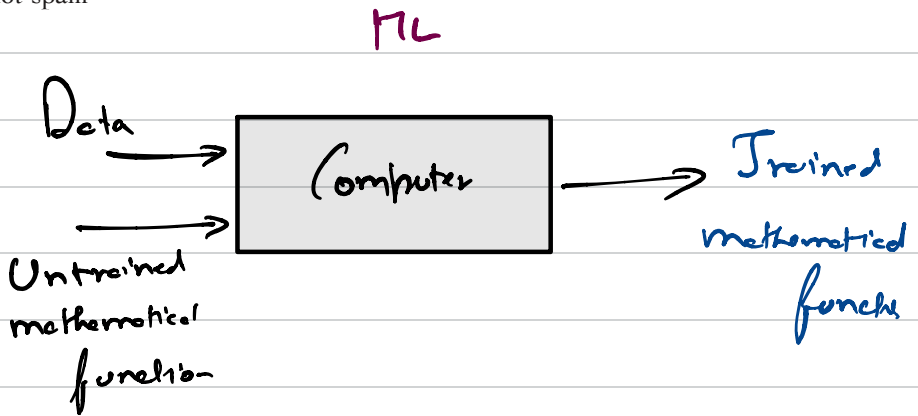


Agenda

1. Basics of ML
2. Linear Models (Implementation) and Gradient Descent Intuition

Arthur Samuel: ML provides a way to learn, without being EXPLICITLY programmed.

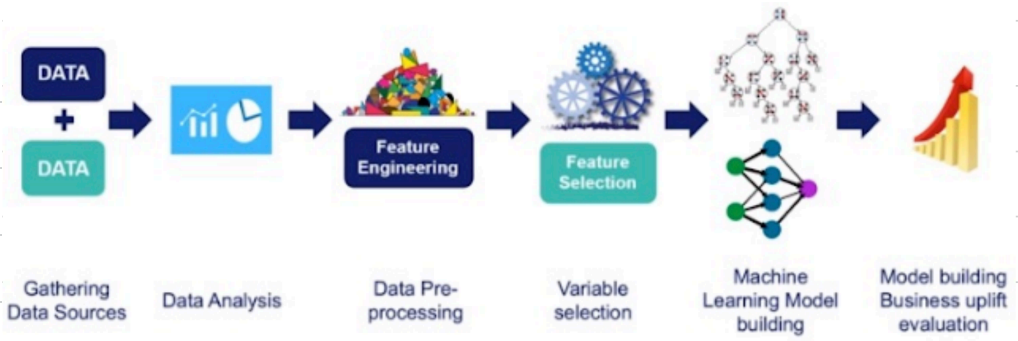
```
for word in email in ['lottery', 'prize', 'money']:  
    return 'spam'  
else:  
    return 'not-spam'
```



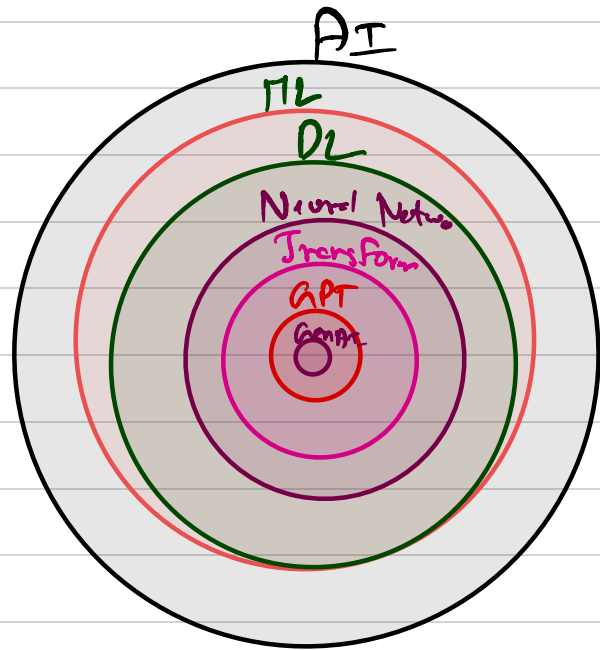
JOE	SALARY
1	10
2	20
3	30
4	40
5	50
6	60

$$\text{Salary} = w \times \text{JOE}$$

$$\text{Salary} = 10 \times \text{JOE}$$



Ht | wt | BMI | Diabetes



Linear Regression

$$w_1 \times f_1 + w_0$$

JOE	SALARY
1	30
2	40
3	50
4	60
5	70
6	80

$$w_1 \times \text{JOE} + w_0$$

w_1 is 10, w_0 is 20

$w_1 \ w_2 \ w_3 \ w_4 \ w_5 \ w_6 \ w_0$
 $f_1 \ f_2 \ f_3 \ f_4 \ f_5 \ f_6 \ y$

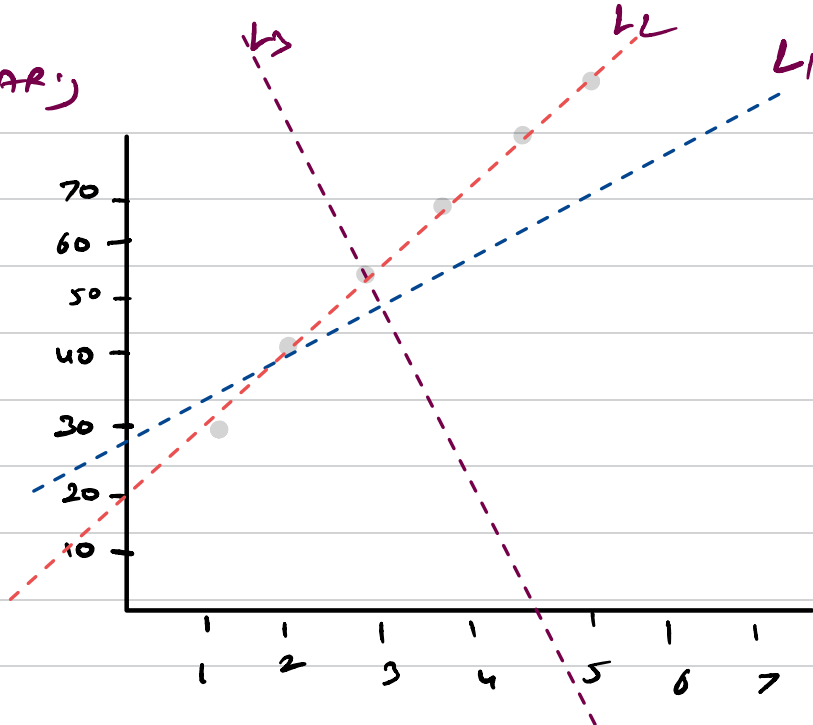
x_1							
x_2							
x_3							
x_4							
x_5							
x_6							
x_7							
x_8							

JOE

SALARY

1
2
3
4
5
6

30
40
50
60
70
80



$$y = w \times \text{Bedrooms} + w_0$$

$$w = 10$$

Bedrooms	Actual rent (₹k)
1	12
1	8
2	22
2	18
3	33
3	27
4	46
4	34
5	55
5	45

$$\hat{y}(w=10)$$

10
10
20
20
30
30
40
40
50
50

$$y - \hat{y}$$

-2
-2
2
-2
3
-3
6
-6
5
-5

$$(y - \hat{y})^2$$

4
4
4
4
9
9
36
36
25
25

Here RMSE tells you, your predictions are about 4 units away from actual.

$$=\sqrt{15.6}$$



3.94968353

Action this item to copy this number

$$MSE = \frac{1}{N} \sum_{i=1}^n (y_i - \hat{y})^2 = 15.6$$

$$Root\ MSE = \sqrt{MSE} = RMSE$$

MAE vs MSE vs RMSE

MAE = Mean absolute error (use it when you're dealing with very small target dataset)

MSE = Mean Squared Error (is de-factor for any kind of continuous numeric prediction model.

RMSE = Root of MSE (when you want interpretable loss-value we do RMSE)

$$w = 0 \quad ; \quad \begin{array}{ccc} \text{Error} & 15.6 & MSE \\ \text{[-10, -9.9, ...]} & \otimes & \text{[0.1, ... 10]} \\ w_1, w_2, w_3 \end{array}$$

$$w_1 = [-10, 10], 0.1 \quad ; \quad 201 \approx 200$$

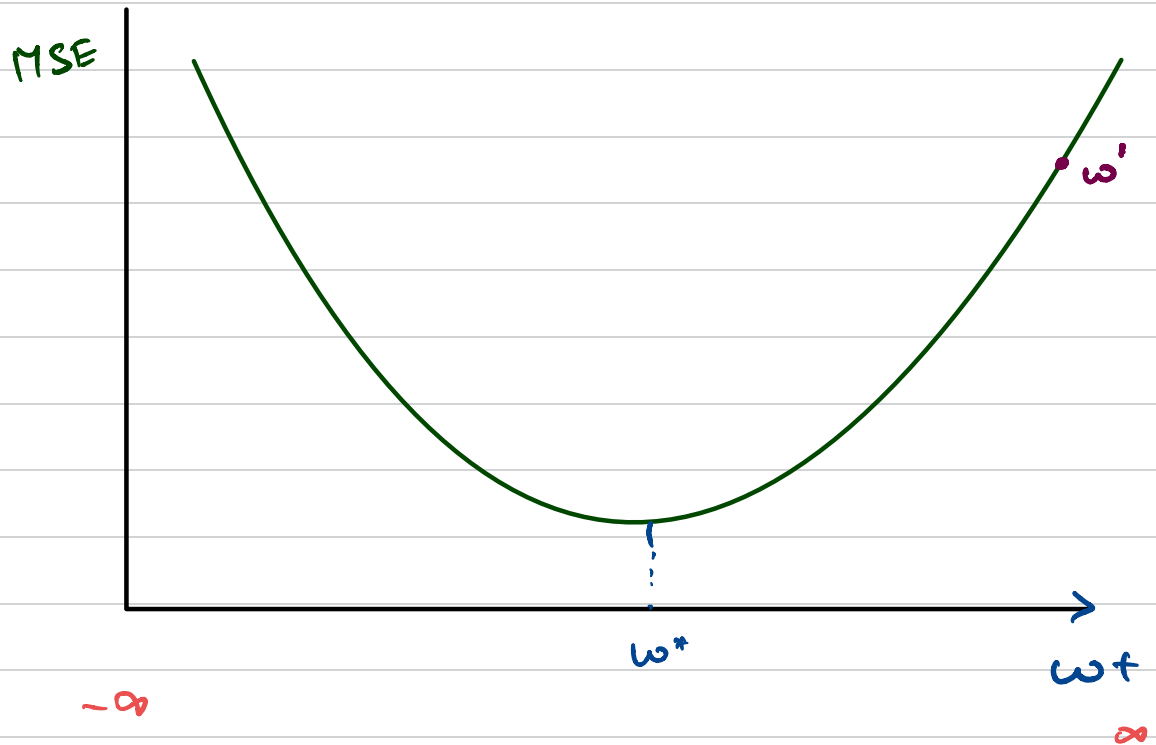
$$w_2 = [-10, 10], 0.1 \quad \approx 200$$

$$w_3 = [-10, 10], 0.1 \quad \approx 200$$

$$\begin{matrix} 1 & 1 & 4 \\ \nearrow & \nearrow & \nearrow \end{matrix}$$

$$MAE = \frac{1}{3} (1 + 1 + 4) = 2$$

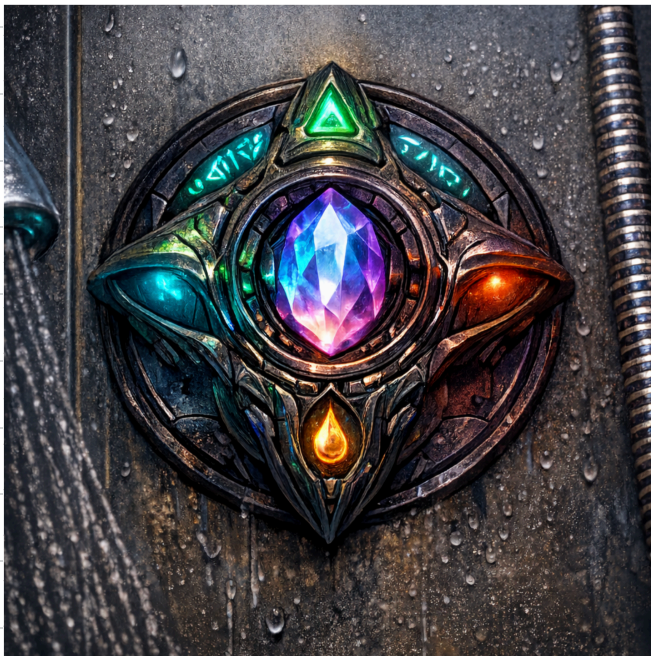
$$RMSE = \sqrt{\frac{1^2 + 1^2 + 4^2}{3}} = 2.45$$

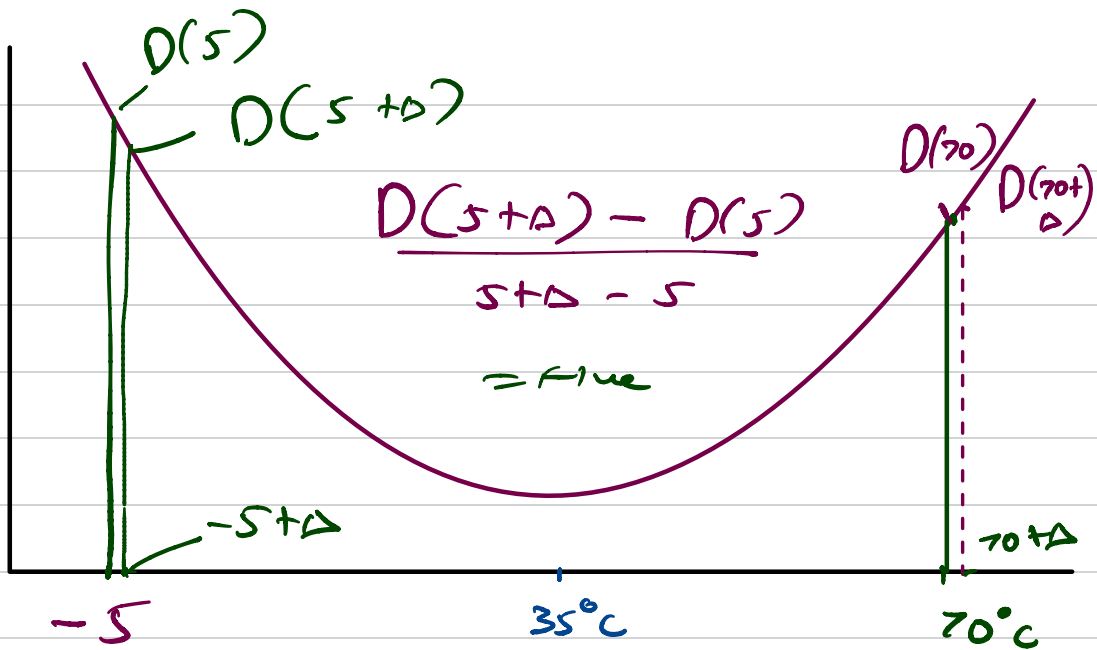


Core ML Workflow

1. **Define:** pick your **features (X)** and **target (y)**, and a baseline.
2. **Prepare data:** collect, clean, and do minimal preprocessing.
3. **Split:** train / validation / test (avoid leakage).
4. **Model:** choose a model family (e.g., linear, tree).
5. **Loss + metric:**
 - **Loss** = what you optimize during training
 - **Metric** = what you report/compare (e.g., MAE/RMSE, F1/AUC)
6. **Train:** fit on train set by minimizing loss.
7. **Evaluate + iterate:** check metric on val/test, fix errors, tune, repeat.

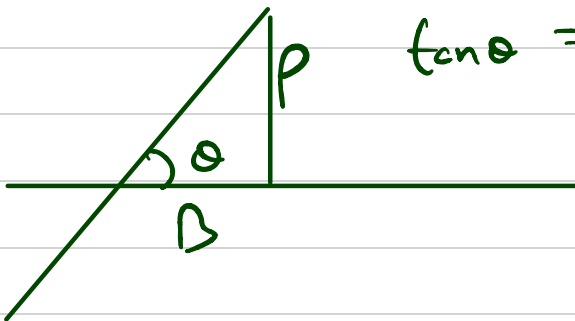
You nudge the random parameter,
'such that loss decreases





$$\text{Slope} = \frac{\text{Rate of change of } y}{\text{Rate of change of } x} = \frac{dy}{dx} = \frac{\Delta y}{\Delta x}$$

$$\frac{y_2 - y_1}{x_2 - x_1} = \frac{y_2 (D(70+\Delta) - D(70))}{x_2 - 70 + \Delta - 70 - x_1}$$



$$\tan \theta = \frac{P}{B} = \frac{y_2 - y_1}{x_2 - x_1}$$

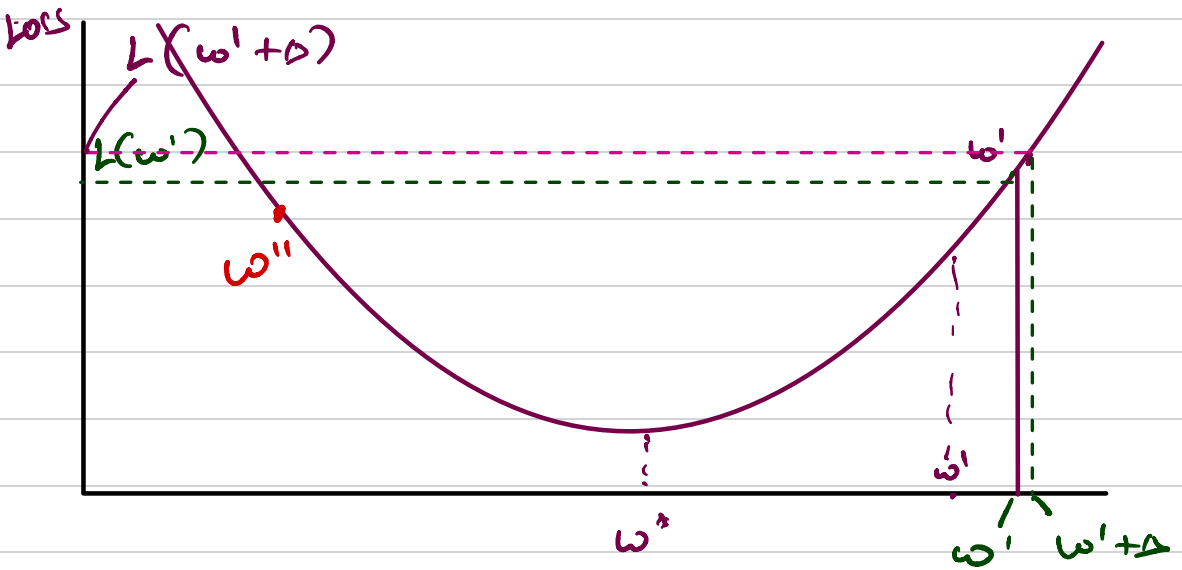
If slope is positive, go to the LEFT

if slope is negative, go to the RIGHT



$$\text{Slope} = \frac{L(\omega' + \Delta) - L(\omega')}{\omega' + \Delta - \omega} = (+) \omega'$$

$$\omega' = \omega' - \text{slope}$$



$$\text{Slope} = \frac{L(w' + \Delta) - L(w')}{w' + \Delta - w'} = \Delta w$$

$$w' = w' - \text{slope} \times \eta \quad \text{step size } 0.001$$

17.4 :

Accuracy METRIC

$$y_{\text{-pred}} = [0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 1 \quad 0 \quad 1 \quad 0]$$
$$y_{\text{-actual}} = [0 \quad 1 \quad 0 \quad 0 \quad 1 \quad 0 \quad 1 \quad 1 \quad 0]$$

✓ ✓ ✓ ✓ ✓ × × ✓ ✓
7/9

$$\text{Accuracy} = \frac{\# \text{ Correct}}{\# \text{ Prediction}}$$

100 emails - 90 - N.S
10 - S } Model → N.S
- 90%

Acc metric

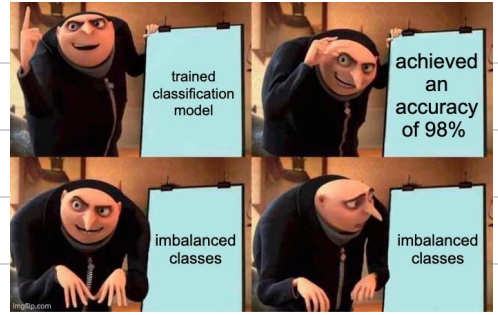
Very mis Leadin For
class imbalance.

My Boss when finds out my
Model is 90% accurate!



Me: Trained my Model
on an imbalanced Dataset
with 90% of one class!





Spam $\rightarrow$ (+)ve Class

Not-Spam $\rightarrow$ (-)ve Class

Actual

Predicted

Spam

Spam

True Positive

Spam

Not Spam

False Negative

Not Spam

Spam

False Positive

Not Spam

Not Spam

True Negative

CONFUSION MATRIX



		Predicted	
		Not spam	Spam
Actual	Not spam	TN	FP
	Spam	FN	TP

Q1. 400 Emails

Actual (360 - N.S.
40 - S

(Model $\rightarrow$ 400 - N.S

Prediction.

$$TN + FN = 400$$

		Predictor	
		Not Spam	Spam
Actual	Not Spam	360 TN	0 FP
	Spam	40 FN	0 TP

Q2. 100 Emails

N.S = 30, S = 70

Predicts 60 - S $\rightarrow$ 50 - S
10 - N.S.
40 - N.S. $\left\{ \begin{array}{l} 20 - S \\ 20 - N.S \end{array} \right.$
FP + TP

		Predictor	
		Not Spam	Spam
Actual	Not Spam	20 TN	10 FP
	Spam	20 FN	50 TP

Gnana Prakash To: Me 10:26 PM

True Negative (TN) = 20
False Positive (FP) = 10
False Negative (FN) = 20
True Positive (TP) = 50

Q.3.

Total Transaction: 10000

Fraud: 200

Not Fraud 9800

M - Fraud 200 $\left(\begin{array}{l} 160 - F \\ 140 - NF \end{array} \right)$

N - Fraud 9700 $\left(\begin{array}{l} 40 - F \\ 9660 - NF \end{array} \right)$

	Predicted	
	Not Spam	Spam
Actual	Not Spam	9660 TN
	Spam	140 FP
Actual	Not Spam	40 FN
	Spam	160 TP

● Ankit Tiwari To: Me 10:31 PM

TP-160 FP-140 TN- 9660 FN-40

$$\text{Precision} = \frac{TP}{TP + FP} = 0.53$$

Precision = Out of all the transactions I predict as fraud, roughly 53% of em are actually fraud.

$$\text{Recall} = \frac{TP}{TP + FN} = 0.8$$

Out of all the transactions that are ACTUALLY fraud, I'm able to identify 80% of em

How to change precision and recall for a given model.

Threshold. prob > 0.5 = Positive, if prob < 0.5 = Negative.

Precision and Recall

Case 1:

1. You incorrectly received an email for donation from nigerian prince.
2. Gmail incorrectly classified a job offer letter from your dream company in your spam and you missed it.

Actual : True

FN

Actual → True

FP

Case 2:

1. Healthy patient was identified as covid positive
2. Covid patient was identified as covid negative.

Actual →

FP

FN

Predictor

Not Spam

Spam

Not
Spam

TN

FP

Actual

Spam

FN

TP

$$\text{Precision} = \frac{TP}{TP + FP}$$

$$\text{Recall} = \frac{TP}{TP + FN}$$

Precision = How good I'm at predicting correctly

Recall = How good am I at identifying all the positive classes present

